TAMILARASAN V

tamilarasanv721@gmail.com | +91 6383048547 | linkedin.com/in/tamil-arasan-v | github.com/tamilarasan2029

Software engineer with end-to-end ownership of production systems in Python and Java — from schema design and backend APIs through data pipelines, CI/CD, and frontend delivery. Experienced building complete applications solo: designing realistic data schemas, implementing multi-stage processing pipelines, and calibrating system behavior against real-world data. Comfortable in unfamiliar technical domains — built a full computer-vision pipeline from research paper to working system, analyzing failures to determine whether issues came from the model, the data, or the pipeline itself. Clear, structured communicator who documents architecture decisions and can explain complex systems simply. Values working software and practical tradeoff over unnecessary complexity.

TECHNICAL SKILLS

Languages	Python,Java, JavaScript, SQL
Backend	Spring Boot, FastAPI, Node.js, RESTAPIs
Frontend	React.js, HTML5, CSS3, JavaScript
Databases& Schema	PostgreSQL, MySQL, Schema Design, Query Optimization, JSON, OCR
Cloud/Storage	AWS S3, Azure (learning), GCP (learning)
Dev Tools	Eclipse, VS Code, SonarQube, soapUI,Git,CI/CP
AI/ML	TensorFlow,OpenCV,FastR-CNN,ImageAI,Deep Learning
Other	Arduino,EmbeddedC,

Experience

Assistant System Engineer — Tata Consultancy Services Apr 2024 – May 2025

● Owned a secure client-admin portal end-to-end (Java, Spring Boot, MySQL) — from schema design through implementation, testing, and production support — as the sole engineer on the Client-Service Mapping module.

● Designed the complete database schema from scratch — relational models, indexes, and constraints — making explicit tradeoffs between normalization and query performance.

● Built the dynamic service-allocation logic for the mapping module, ensuring real-time data consistency across backend services under changing business rules.

● Diagnosed and fixed a backend performance bottleneck by analyzing query execution plans and restructuring indexes, cutting processing time by 40% — root-caused the issue to missing composite indexes rather than application logic.

● Built and validated REST APIs with Spring Boot and soapUI; integrated automated builds and deployments into a CI/CD pipeline,and used SonarQube to maintain code quality across the module.

● Worked with incomplete or evolving requirements across the SDLC, clarifying scope directly with stakeholdersand documenting decisions for future maintainers.

Full-Stack Developer - freelancer

Jun 2025 - May 2026

● Owned a full-stack hospital management platform end to end — from initial specification through architecture, implementation, testing, and delivery — as the sole engineer, delivering the entire application independently with complete ownership of every technical decision.

● Designed the system architecture across three services: a Python (FastAPI) backend for data ingestion and access control, a Node.js middleware layer for async event processing, and a React frontend — deciding where business logic belonged as requirements changed mid-build.

● Made explicit architecture tradeoffs: chose a separate Node.js middleware layer over handling notifications directly in FastAPI, prioritizing decoupled async processing over simplicity, and would reconsider this split today given the added operational complexity.

● Built role-based access control (admin, doctor, receptionist) and secure document storage in FastAPI, with input validation and structured error handling to keep the system reliable without a dedicated QA process.

● Debugged race conditions and data-consistency failures between asynchronous notification events and the primary database — root-caused issues to event-ordering assumptions rather than the database layer itself.

● Designed a normalized PostgreSQL schema covering patients, appointments, prescriptions, and billing from scratch; wrote and optimized complex multi-table joins, materially improving reporting query performance.

● Built the React frontend to consume and parse nested JSON APIs across all modules, handling schema changes and edge cases in the data shape as the backend evolved.

● Set up a CI/CD pipeline to automate testing and deployment across the FastAPI, Node.js, and React services,reducing manual release effort and catching integration issues earlier.

● Tested the system incrementally at each layer — API-level validation, database constraint checks, and manual end-to-end flows — in the absence of a formal test suite,and can speak to what a proper testing strategy would look like in hindsight.

● Managed the full project lifecycle independently with no direct supervision — clarifying ambiguous requirements,scoping features, making architectural calls under time pressure, and carrying the system through to a working,deployed product.

Academy Project

CAM-Based Object Detection & Identification Python · OpenCV · TensorFlow · Fast R-CNN · PostgreSQL
Oct 2022 – Apr 2023

- Designed and implemented a real-time object detection system using Fast R-CNN and OpenCV, capable of detecting and classifying multiple objects in images with bounding box annotations.
- Built the full pipeline in Python: image segmentation → feature extraction (Hessian-Laplace + PCA-SIFT) → ROI pooling → object classification using a Faster-RCNN convolutional neural network.
- Integrated TensorFlow and Image AI deep learning libraries; trained the model on benchmark datasets (COIL100, Caltech 101, ETH80, MNIST) achieving high recall and precision rates.
- Used MySQL (PGAdmin) as the backend data store for object metadata and detection logs — directly applicable to document and data pipeline workflows.

EDUCATION

Karpagam College of Engineering — B.E., Electronics & Communication Engineering | Aug 2019 – Apr 2023 Coimbatore, India | CGPA: 7.85